

```
In [ ]: #Positional Arguments
```

```
In [1]: #1) Add two numbers
```

```
def add(a, b):  
    return a + b
```

```
print(add(5, 3))
```

8

```
In [2]: #2. Calculate area of a rectangle
```

```
def area(length, width):  
    return length * width
```

```
print(area(10, 5))
```

50

```
In [3]: #3. Greet a person
```

```
def greet(name, age):  
    print(f"Hello {name}, you are {age} years old.")
```

```
greet("Rashmi", 25)
```

Hello Rashmi, you are 25 years old.

```
In [4]: #4. Multiply three numbers
```

```
def multiply(x, y, z):  
    return x * y * z
```

```
print(multiply(2, 3, 4))
```

24

```
In [5]: #5. Find simple interest
```

```
def simple_interest(p, r, t):  
    return (p * r * t) / 100
```

```
print(simple_interest(1000, 5, 2))
```

100.0

```
In [6]: #Display student details
def student(name, roll, branch):
    print(name, roll, branch)

student("Sid", 23, "CSE")
```

Sid 23 CSE

```
In [7]: #7. Power function
def power(base, exponent):
    return base ** exponent

print(power(3, 4))
```

81

```
In [8]: #8. Convert temperature
def to_fahrenheit(celsius):
    return (celsius * 9/5) + 32

print(to_fahrenheit(30))
```

86.0

```
In [9]: #9. Distance formula
def distance(x1, y1, x2, y2):
    return ((x2-x1)**2 + (y2-y1)**2) ** 0.5

print(distance(0, 0, 3, 4))
```

5.0

```
In [10]: #10. Create a full name
def full_name(first, middle, last):
    return f"{first} {middle} {last}"

print(full_name("Rashmi", "S.", "Balurkar"))
```

Rashmi S. Balurkar

```
In [ ]: #Keyword Argumens
```

```
In [11]: #1. Greet a person
def greet(name, age):
    print(f"Hello {name}, age {age}")

greet(age=25, name="Rashmi")
```

Hello Rashmi, age 25

```
In [12]: #2. Display product details
def product(name, price, brand):
    print(name, price, brand)

product(price=500, name="Shoes", brand="Puma")
```

Shoes 500 Puma

```
In [13]: #3. Calculate simple interest
def simple_interest(p, r, t):
    return (p * r * t) / 100

print(simple_interest(t=2, p=1000, r=5))
```

100.0

```
In [14]: #4. Area of triangle
def triangle_area(base, height):
    return 0.5 * base * height

print(triangle_area(height=10, base=6))
```

30.0

```
In [15]: #5. Student details
def student(name, roll, marks):
    print(f"Name={name}, Roll={roll}, Marks={marks}")

student(roll=21, marks=88, name="Rashmi")
```

Name=Rashmi, Roll=21, Marks=88

```
In [16]: #6. Concatenate strings
def join_strings(a, b, c):
    return a + b + c
```

```
print(join_strings(c="!", a="Hello ", b="World"))
```

Hello World!

```
In [17]: #7. Power function
def power(base, exponent):
    return base ** exponent

print(power(exponent=3, base=2))
```

8

```
In [18]: #8. BMI calculation
def bmi(weight, height):
    return weight / (height ** 2)

print(bmi(height=1.75, weight=70))
```

22.857142857142858

```
In [19]: #9. Create email
def email(username, domain):
    return f"{username}@{domain}"

print(email(domain="gmail.com", username="rashmi"))
```

rashmi@gmail.com

```
In [20]: #10. Travel cost
def travel_cost(distance, rate):
    return distance * rate

print(travel_cost(rate=10, distance=120))
```

1200

```
In [ ]: #Default Argument
```

```
In [21]: #1. Greeting with default message
def greet(name="Guest"):
    print(f"Hello, {name}!")
```

```
greet()
greet("Rashmi")
```

Hello, Guest!

Hello, Rashmi!

In [22]: *#2. Power function with default exponent*

```
def power(base, exponent=2):
    return base ** exponent
```

```
print(power(5))
print(power(5, 3))
```

25

125

In [23]: *#3. Area of rectangle (default width = 1)*

```
def area(length, width=1):
    return length * width
```

```
print(area(10))
print(area(10, 5))
```

10

50

In [24]: *#4. Student details with default marks*

```
def student(name, roll, marks=0):
    print(name, roll, marks)
```

```
student("Sid", 21)
student("Sid", 21, 88)
```

Sid 21 0

Sid 21 88

In [25]: *#5. Welcome message with default city*

```
def welcome(name, city="Bangalore"):
    print(f"Welcome {name} to {city}!")
```

```
welcome("Rashmi")
welcome("Rashmi", "Mumbai")
```

Welcome Rashmi to Bangalore!

Welcome Rashmi to Mumbai!

In [26]: *#6. Calculate salary with default bonus*

```
def salary(basic, bonus=5000):  
    return basic + bonus
```

```
print(salary(30000))  
print(salary(30000, 8000))
```

35000

38000

In [27]: *#7. Shipping cost with default rate*

```
def shipping_cost(weight, rate=10):  
    return weight * rate
```

```
print(shipping_cost(5))  
print(shipping_cost(5, 15))
```

50

75

In [29]: *#8. Make a full name (default middle name)*

```
def full_name(first, last, middle=""):  
    return f"{first} {middle} {last}".strip()
```

```
print(full_name("Rashmi", "Balurkar"))  
print(full_name("Rashmi", "Balurkar", "S."))
```

Rashmi Balurkar

Rashmi S. Balurkar

In [30]: *#9. Display product (default discount)*

```
def product(name, price, discount=0):  
    return price - discount
```

```
print(product("Shoes", 1000))  
print(product("Shoes", 1000, 100))
```

1000

900

In [31]: *#10. Calculate simple interest (default rate = 5%)*

```
def simple_interest(p, t, r=5):  
    return (p * t * r) / 100  
  
print(simple_interest(1000, 2))  
print(simple_interest(1000, 2, 7))
```

100.0

140.0

In []: *#2nd Dec 2025*

In []: *#Variable Length Argument*

In [32]: *#1. Sum of numbers*

```
def total_sum(*numbers):  
    return sum(numbers)  
  
print(total_sum(1, 2, 3, 4))
```

10

In [33]: *#2. Multiply any number of values*

```
def multiply_all(*values):  
    result = 1  
    for v in values:  
        result *= v  
    return result  
  
print(multiply_all(2, 3, 4))
```

24

In [34]: *#3. Print student names*

```
def students(*names):  
    for n in names:  
        print(n)  
  
students("Amit", "Rashmi", "John")
```

Amit
Rashmi
John

```
In [35]: #4. Find maximum
def max_value(*nums):
    return max(nums)

print(max_value(5, 20, 10))
```

20

```
In [36]: #5. Concatenate strings
def join_strings(*words):
    return " ".join(words)

print(join_strings("Python", "is", "fun"))
```

Python is fun

```
In [ ]: #Keyword variable Length arguments(kwargs)
```

```
In [37]: #1. Display person details
def person_details(**info):
    print(info)

person_details(name="Rashmi", age=25, city="Bangalore")

{'name': 'Rashmi', 'age': 25, 'city': 'Bangalore'}
```

```
In [38]: #2. Build a dictionary
def build_dict(**pairs):
    return pairs

print(build_dict(a=1, b=2, c=3))

{'a': 1, 'b': 2, 'c': 3}
```

```
In [39]: #3. Student marks
def marks(**subject_marks):
    for subject, marks in subject_marks.items():
        print(subject, "=", marks)
```



```
marks(Math=90, Science=85, English=88)
```

```
Math = 90  
Science = 85  
English = 88
```

```
In [ ]: #Using both *args and **kwargs
```

```
In [40]: #1. Mixed arguments  
def display(*args, **kwargs):  
    print("Args:", args)  
    print("Kwargs:", kwargs)  
  
display(1, 2, 3, name="Rashmi", age=25)
```

```
Args: (1, 2, 3)  
Kwargs: {'name': 'Rashmi', 'age': 25}
```

```
In [41]: #2. Shopping bill  
def shopping(*items, **prices):  
    print("Items:", items)  
    print("Prices:", prices)  
  
shopping("Milk", "Eggs", bread=40, rice=60)
```

```
Items: ('Milk', 'Eggs')  
Prices: {'bread': 40, 'rice': 60}
```

```
In [ ]:
```